Day 132 | Unit 09 | Teacher plan

### Display, Describe, and Summarize Data

# Summarize Data in Context

Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d

Learning target: I can write a statistical summary tied to its context.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

## 0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

## 5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A useful summary names the group and attribute, units, observation count, center, variability, and shape. Explain why the chosen center and spread fit the data and mention any striking value.

## 17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

## 42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

## Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

## Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references:

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## 01 / Learn

A useful summary names the group and attribute, units, observation count, center, variability, and shape. Explain why the chosen center and spread fit the data and mention any striking value.

## Worked example 1

Reading times in minutes are 8, 9, 9, 10, 10, 11. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

1. A useful summary names the group and attribute, units, observation count, center, variability, and shape.

2. Explain why the chosen center and spread fit the data and mention any striking value.

3. Calculate the center and spread before interpreting them in context.

4. Other accurate, justified summaries are acceptable.

5. State the result in the form requested: 6 observations of reading time in minutes. Mean 19/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable..

Answer: 6 observations of reading time in minutes. Mean 19/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

## Worked example 2

Six students recorded reading times (minutes): 2, 4, 4, 6, 6, 8. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

1. A useful summary names the group and attribute, units, observation count, center, variability, and shape.

2. Order the observations.

3. Middle positions give median 5 and half medians give IQR 2.

4. Mean is 30/6

5. averaging distances gives MAD 5/3.

6. Tie the chosen summary to the displayed shape and context.

Answer: 6 observations; minutes; median 5, IQR 2; mean 5, MAD 5/3. Roughly balanced; mean/MAD or median/IQR with justification.

# Day 132 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

### 01 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 4, 5. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

### 02 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

### 03 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 7, 8. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

### 04 / Constructed response

Reading times in minutes are 6, 10, 10, 14, 14, 18. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

### 05 / Constructed response

Reading times in minutes are 2, 8, 8, 14, 14, 20. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

## 06 / Constructed response

Reading times in minutes are 3, 9, 9, 15, 15, 21. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

## 07 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 5, 14. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

## 08 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 3, 6, 6, 9, 9, 12. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

## 09 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 5, 9, 9, 13, 13, 25. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

## 10 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 11, 20. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

Day 132 | Unit 09 | Answer key

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Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d

Learning target: I can write a statistical summary tied to its context.

### 1. 6 observations of reading time in minutes. Mean 7/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

### 2. 6 observations of reading time in minutes. Mean 11/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

### 3. 6 observations of reading time in minutes. Mean 13/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

### 4. 6 observations of reading time in minutes. Mean 12 min, MAD 10/3 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

### 5. 6 observations of reading time in minutes. Mean 11 min, MAD 5 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

### 6. 6 observations of reading time in minutes. Mean 12 min, MAD 5 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

### 7. 6 observations of reading time in minutes. Median 7/2 min, IQR 2 min; right-skewed with a high value 14 min, so median/IQR resist that value.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

### 8. 6 observations; minutes; median 15/2, IQR 3; mean 15/2, MAD 5/2. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 15/2 and half medians give IQR 3. Mean is 45/6; averaging distances gives MAD 5/2. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation

### 9. 6 observations; minutes; median 11, IQR 4; mean 37/3, MAD 14/3. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 11 and half medians give IQR 4. Mean is 74/6; averaging distances gives MAD 14/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation

### 10. 6 observations of reading time in minutes. Median 19/2 min, IQR 2 min; right-skewed with a high value 20 min, so median/IQR resist that value.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response